



## Koneru Lakshmaiah Education Foundation

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### OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

#### Project Problem Statement Submission Form

##### Section / Team Details

Section No: 03

Team No: 15

Project Title: Hospital Emergency Patient Monitoring System Using Linux IPC

##### Team Members

| Roll Number | Student Name      | Signature |
|-------------|-------------------|-----------|
| 2520030471  | SAI ANANYA ROY    |           |
| 2520030504  | DEVUNI SHREENITHA |           |

### 1. Abstract

The **Hospital Emergency Patient Monitoring System** is a Linux-based project that demonstrates **Inter-Process Communication (IPC)**. It simulates patient monitoring, emergency processing, and doctor alerts using multiple processes. The system takes the **Patient ID, heart rate, blood pressure, temperature, and oxygen level**, sends the patient data through a pipe, checks the patient's health status, and sends the result through a FIFO. The system classifies the patient as **NORMAL, WARNING, or EMERGENCY** and displays the **condition, recommended action, and doctor alert**.

### 2. Problem Statement

The Hospital Emergency Patient Monitoring System is designed to collect patient details such as heart rate, blood pressure, temperature, and oxygen level. It uses Linux pipes and FIFOs to send patient data between processes, check the patient's health condition, and display a NORMAL, WARNING, or EMERGENCY status with the recommended action and alert information.

### 3. Objectives

- To simulate a Hospital Emergency Patient Monitoring System.
- To demonstrate Inter-Process Communication (IPC) in Linux.
- To use pipes and FIFOs for communication between processes.
- To collect and transfer Patient ID, heart rate, blood pressure, temperature, and oxygen level.
- To check patient vital signs and classify the condition as NORMAL, WARNING, or EMERGENCY.
- To display the condition, recommended action, and doctor alert information.

#### 4. Proposed Methodology

\* **Patient Monitoring** – Takes the Patient ID and heart rate from the user and sends the data through a pipe.

\* **Emergency Processor** – Receives the patient data, checks the heart rate, and gives a status such as **LOW, NORMAL, WARNING, or EMERGENCY**. It also displays the simulated condition and recommended action.

\* **Doctor Alert** – Receives the processed information through a **FIFO** and displays the emergency alert.

##### Flow:

Patient Monitoring → Pipe → Emergency Processor → FIFO → Doctor Alert

#### 5. Operating Systems Concepts / Linux APIs Used

| OS Concept / Linux API / System Call | Purpose in the Project                  |
|--------------------------------------|-----------------------------------------|
| fork()                               | Creates processes                       |
| pipe()                               | Communication between related processes |
| mkfifo()                             | Creates a named pipe                    |
| read()                               | Receives data                           |
| write()                              | Sends data                              |
| open() / close()                     | Manages communication channels          |
| waitpid()                            | Process synchronization                 |

|                             |                                                               |
|-----------------------------|---------------------------------------------------------------|
| IPC                         | Enables process communication                                 |
| Process Synchronization     | Coordinates processes                                         |
| File Descriptors<br>execl() | Accesses pipes and FIFOs<br>Executes the Doctor Alert program |
| unlink()                    | Removes the FIFO after use                                    |

## 6. Individual Contribution

| Roll Number | Student Name      | Individual Responsibility |
|-------------|-------------------|---------------------------|
| 2520030471  | SAI ANANYA ROY    | Abstract                  |
| 2520030504  | DEVUNI SHREENITHA | Code Implementation       |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                       |
|----------------------------|-------------------------------|
| Linux / Ubuntu             | Project execution environment |
| C / C++                    | Program implementation        |
| GCC                        | Compilation                   |
| POSIX APIs                 | System and IPC functions      |
| VS Code/                   | Compilation and testing       |
| Linux Terminal             | Code development              |

## 8. Expected Outcome

The system will collect and process **patient vital signs**, transfer the patient data using **pipes and FIFOs**, and classify the patient's condition as **NORMAL, WARNING, or EMERGENCY**. It will display the **condition, recommended action, and doctor alert information**, demonstrating the use of **Linux processes and Inter-Process Communication (IPC)**.

## 9. GitHub Repository

**GitHub Repository Link:** [https://github.com/shreenithadevuni/Linux-Inter-Process-Communication-Using-Pipes-and-FIFOs-\\_OSSP](https://github.com/shreenithadevuni/Linux-Inter-Process-Communication-Using-Pipes-and-FIFOs-_OSSP)

**Repository Created:** \_\_\_\_\_

☐ Yes   ☐ No

## Faculty Name and Signature

**Faculty Name:** \_\_\_\_\_

**Signature:** \_\_\_\_\_

**Remarks:** \_\_\_\_\_

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## Faculty Use

**Project Approved:** \_\_\_\_\_

☐ Yes   ☐ No

**Date:** \_\_\_\_\_

**Faculty Signature:** \_\_\_\_\_